Markov Chains: From Theory to Applications

An IASc Summer Research Report

By

Samriddha Ganguly
BS-MS Physics
Indian Institute of Science Education & Research, Bhopal

On work carried out at the Indian Institute of Science under the guidance of

Dr. Arvind Ayyer
Department of Mathematics
Indian Institute of Science, Bengaluru.

Contents

1 Theoretical Premise of Markov Chains **4**
1.1 Defining a Markov Chain 4
1.2 Chapman Kolmogorov Equations 5
1.3 Class structure & Class Properties 6
1.4 Hitting Times and Probabilities 9
1.5 Long-run Proportions of time & Stationary Distributions 11
1.6 Convergence to Equilibrium, Limiting Theorems and Ergodicity 11
1.7 Time Reversal 13
1.8 The Poisson Process 14
1.9 A Brief Introduction to Continuous Time Markov Chains 18

2 Applications of Markov Chain Modelling **20**
2.1 Ultimate Instability of Aloha Protocol 20
2.2 The Gambler's Ruin 22

3 Markov Chain Monte Carlo Methods **24**
3.1 Introduction 24
3.2 Metropolis-Hastings Algorithm & Gibbs Sampler 24
3.3 Applications in the Ising Model 26

4 Brownian Motion & Wiener Processes **29**

5 Prospective Future Research & Conclusion **31**

6 Appendix **32**
6.1 Appendix 1: Historical Aspect 32
6.2 Appendix 2: Codes 33

References **38**

Acknowledgement

I would like to firstly thank my guide, Dr. Arvind Ayyer for his wonderful guidance as well as useful insights. The time he dedicated was fruitful for me to learn a lot of new things during the course of this internship. I would also like to thank all the three Science Academies and all organizers of the SRFP Program for giving me this wonderful opportunity to work at an institute like IISC, Bengaluru. A lot of thanks to my friend and companion, Surya from IISER Trivandrum for the numerous presentations, discussions and interactions we had during my internship.

I would like to thank all the great scientists and professors whose resources I have utilized and particularly I would like to thank Dr. Prahlad Vaidyanathan who taught me the basics of probability at IISER Bhopal. I would also like to thank Dr. Dheeraj Kulkarni who graciously agreed to give me a letter of recommendation for this internship.

Lastly, I would extend my gratitude towards my parents without whose support none of this would be possible.

Abstract

This project aims to give a theoretical background to Markov Processes with particular emphasis on how we can use them to model various systems.

I have presented here all the theoretical premise I gathered during my presentations with Dr. Ayyer over the summer. After the theory I looked into applications and simulations of Markov chains and applied them to a classical model of Statistical Physics: the Ising model.

This project additionally includes codes, future open research questions and historical background of Markov chains along with a short introduction to Brownian Motion.

Chapter 1

Theoretical Premise of Markov Chains

1.1 Defining a Markov Chain

In many real life situations we notice that often phenomena we observe are in fact not truly independent. Let us take the example of weather forecasting. We cannot say that weather of one day is totally independent of the previous days. However, we can say that the conditions of the previous day may have some bearing on the next day. This is essentially the idea behind Markov chains. Now, we will try to define this mathematically.

Definition: A stochastic process $\{X_n\}_{n\in\mathbb{N}_0}$ taking values in a countable state space S is called a Markov chain (or said to have the Markov property) if

$$\begin{aligned} P[X_{n+1} = i_{n+1}|X_n = i_n, X_{n-1} = i_{n-1}, ..., X_1 = i_1, X_0 = i_0] \\ = P[X_{n+1} = i_{n+1}|X_n = i_n] \end{aligned} \tag{1.1}$$

So basically we can interpret (1.1) as: the conditional distribution of any future state X_{n+1}, given the past states $X_0, X_1, ...X_{n-1}$ and present state X_n is independent of the past states and is solely dependent on the present state. Now let P_{ij} denote the probability that the process will when in state i, next make a transition to state j. Since probabilities are non-negative and since the process must make a transition into some state we have that:

$$P_{ij} \geq 0, \quad i, j \geq 0; \quad \sum_{j=0}^{\infty} P_{ij} = 1, \quad i = 0, 1, ...$$

Let P denote the matrix of one-step transition probabilities P_{ij}, so that

$$P = \begin{pmatrix} P_{00} & P_{01} & P_{02} & \dots \\ P_{10} & P_{11} & P_{12} & \dots \\ \vdots & & & \\ P_{i0} & P_{i1} & P_{i2} & \dots \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix}$$

Such an array is called a The one-step transition matrix. Knowledge of the transition probability matrix and of the distribution of X_0 enables us, in theory, to compute all probabilities of interest. [2]

1.2 Chapman Kolmogorov Equations

Having defined the one-step transition probability, let us move forward with our logic and try to define the two-step transition probability. We can also define the two-stage transition probability P_{ij}^2 that a system presently in state i will be in state j after two additional transitions. That is,

$$P_{ij}^2 = P(X_{m+2} = j | X_m = i)$$

The P_{ij}^2 can be computed from the P_{ij} as follows:

$$\begin{aligned} P_{ij}^2 &= P(X_2 = j | X_0 = i) \\ &= \sum_{k \in A} P(X_2 = j, X_1 = k | X_0 = i) \\ &= \sum_{k \in A} P(X_2 = j | X_1 = k, X_0 = i) P(X_1 = k | X_0 = i) \\ &= \sum_{k \in A} P_{kj} P_{ik} \end{aligned}$$

In general, we define the $n-$step transition probabilities, P_{ij}^n as:

$$P_{ij}^n = P(X_{m+n} = j | X_m = i)$$

Now, we will state the Chapman-Kolmogorov equations and prove it using our current knowledge.

Proposition 1.

$$P_{ij}^n = \sum_{k \in A} P_{ik}^{(r)} P_{kj}^{(n-r)} \tag{1.2}$$

Where A is just the state space, $\forall\ i, j \in A$ and $0 < r < n$.

Proof.

$$\begin{aligned} P_{ij}^n &= P(X_n = j | X_0 = i) \\ &= \sum_{k \in A} P(X_n = j, X_r = k | X_0 = i) \\ &= \sum_{k \in A} P(X_n = j | X_r = k, X_0 = i) P(X_r = k | X_0 = i) \\ &= \sum_{k \in A} P_{kj}^{(n-r)} P_{ik}^{(r)} \end{aligned}$$

□

This can be validated by the drawing analogy from the law of total probability. By this equation of taking an intermediate we are accounting for one partition given by A_k, however we know every partition will also map all the event that is going from i to j just in different ways, i.e. their probability will be same. Hence we conclude this procedure is valid. Also we see from linear algebra that this is nothing but the matrix multiplication of the n step matrices. Thus we can eventually state if $P^{(n)}$ denote the matrix of n-step transition probabilities P_{ij}^n, then we can see from (1.2), that:

$$P^{(n)} = P^n$$

i.e. the n-step transition probability matrix can easily be obtained by multiplying the transition matrix P n times.

1.3 Class structure & Class Properties

We will now try to see how we can divide states of a Markov Chains into various "classes". This is useful as members of such classes share certain important properties which are essential for our understanding of Markov processes. Firstly let us state what we mean by accessibility of states. A state j is said to be accessible from a state i iff for some $n \geq 0$ we have:

$$P_{ij}^n > 0.$$

This is denoted by $i \to j$. Now we can state that if both $i \to j$ and $j \to i$ then i and j *communicate* and is denoted by $i \leftrightarrow j$. Now we will state some properties of this communication relation:

i) $i \leftrightarrow i$, $\forall i \geq 0$ (Reflexive)
ii) If $i \leftrightarrow j$ then, $j \leftrightarrow i$ (Symmetric)
iii)If $i \leftrightarrow j$ and $j \leftrightarrow k$, then $i \leftrightarrow k$ (Transitive)

We can see that (i) and (ii) follow easily by definition and since $P_{ii}^0 = 1$, $\forall\ i \in A$. For (iii), we will prove this easily. From given facts we have that $P_{ij}^m > 0$ and $P_{jk}^n > 0$, for some $m, n \geq 0$. Then we can say by (1.2), that:

$$P_{ik}^{n+m} = \sum_{r=0}^{\infty} P_{ir}^m P_{rk}^n \geq P_{ij}^m P_{jk}^n > 0.$$

Hence, we say that $i \to k$ and by same logic $k \to i$. Thus we have proved that $i \leftrightarrow k$.
We can see that this is basically an equivalence relation and thus we can "partition" the state space into various equivalence classes. As a consequence of these relations we can also say two classes are either disjoint or identical. The Markov chain is said to be *irreducible* if there is only one class. [4]

Now let us define the concept of **transience** and **recurrence**. Let f_i denote the probability that starting in state i, the chain will ever return to state i. A state is said to be recurrent if $f_i = 1$ and transient if $f_i < 1$.
On pondering on these definitions we see that in fact, it is an almost sure event that if i is recurrent it will eventually re-enter i that too, infinitely often.
However, for the transient state we can see that each time it leaves a state it has $1 - f_i$ probability of never returning again. . Therefore, starting in state i, the probability that the process will be in state i for exactly n time periods equals $f_i^{n-1}(1 - f_i), n \geq 1$. In other words, if state i is transient then, starting in state i, the number of time periods that the process will be in state i has a geometric distribution with finite mean $1/(1 - f_i)$.

Proposition 2. *State i is*
i) recurrent if $\sum_{n=1}^{\infty} P_{ii}^n = \infty$
ii) transient if $\sum_{n=1}^{\infty} P_{ii}^n < \infty$

The argument leading to the preceding proposition is doubly important because it also shows that a transient state will only be visited a finite number of times (hence the name transient). This leads to the conclusion that in a finite-state Markov chain not all states can be transient. To see this, suppose the states are $0, 1, ..., M$ and suppose that they are

all transient. Then after a finite amount of time (say, after time T_0) state 0 will never be visited, and after a time (say, T_1), state 1 will never be visited, and after a time (say, T_2) state 2 will never be visited, and so on. Thus, after a finite time $T = max\{T_0, T_1, ..., T_M\}$ no states will be visited. But as the process must be in some state after time T we arrive at a contradiction, which shows that at least one of the states must be recurrent.[1]

We can now state that if C is a class and if $i \in C$, $i \to j$, imply $j \in C$, i.e. if the class is not *escapable* , it is called a *closed class*. A state i is called an *absorbing state* if, $\{i\}$ is a closed class.

No we can show that recurrence is infact a class property. For this, let us consider the proposition:

Proposition 3. *If state i is recurrent, and state i communicates with state j, then state j is recurrent.*

Proof. We can consider that $\exists\ n, k, m \in \mathbb{N}_0$ such that, $p_{ij}^n > 0$, $p_{ji}^k > 0$ and $p_{ii}^m > 0$, since $i \leftrightarrow j$. Thus now we can say:

$$p_{jj}^{n+m+k} \geq p_{ji}^k p_{ii}^m p_{ij}^n$$

This follows since the left side of the preceding is the probability of going from j to j in $m+n+k$ steps, while the right side is the probability of going from j to j in $m+n+k$ steps via a path that goes from j to i in k steps, then from i to i in an additional m steps, then from i to j in an additional n steps. From the preceding we obtain, by summing over m, that:

$$\sum_{m=1}^{\infty} p_{jj}^{n+m+k} \geq p_{ji}^k p_{ij}^n \sum_{m=1}^{\infty} p_{ii}^m = \infty$$

Thus it follows from Proposition 2 that indeed j is also recurrent. □

Thus we have shown recurrence and transience are class properties. Also, Proposition 3 along with our previous result that not all states in a finite Markov chain can be transient leads to the conclusion that all states of a finite irreducible Markov chain are recurrent.[1]

Recurrence and Transience of Random Walks

In this section I will like to mathematically show how we can prove the recurrent nature of one-dimensional symmetric random walks and also show how that is not the case for any symmetric random walk in a three-dimensional lattice.

A symmetric random walk on $\mathbb{Z}$: To define a simple random walk on $\mathbb{Z}$, suppose we are in i, then the chain can move to $i-1$ with probability $q = 1-p$ and to $i+1$ with probability p. Here $p, q \geq 0$ and ≤ 1. Suppose we start at O. It is clear that we cannot return to 0 after an odd number of steps, so $p_{00}^{(2n+1)}$ for all n. Any given sequence of steps of length $2n$ from 0 to 0 occurs with probability $p^n q^n$, there being n steps on either side, and the number of such sequences is the number of ways of choosing the n steps on one side from $2n$. Thus

$$p_{00}^{(2n)} = \binom{2n}{n} p^n q^n$$

Stirling's formula provides a good approximation to $n!$ for large n: it is known that as $n \to \infty$:

$$n! \sim \sqrt{2\pi n}(n/e)^n$$

For the n-step transition probabilities we obtain:

$$p_{00}^{(2n)} = \frac{(2n)!}{(n!\,)^2} p^n q^n \sim \frac{(4pq)^n}{A\sqrt{n/2}}$$

Here A is a positive real constant. This is valid for $n \to \infty$. In the symmetric case $p = q = 1/2$, so $4pq = 1$; then for some N and all $n \geq N$ we have:

$$p_{00}^{(2n)} \geq \frac{1}{2A\sqrt{n}}$$

So, by summation:

$$\sum_{n=N}^{\infty} p_{00}^{(2n)} \geq \frac{1}{2A} \sum_{n=N}^{\infty} \frac{1}{\sqrt{n}} = \infty$$

which shows that the random walk is recurrent. On the other hand, if $p \neq q$ then$4pq = r < 1$, so by a similar argument, for some N,

$$\sum_{n=N}^{\infty} p_{00}^{(n)} \leq \frac{1}{A} \sum_{n=N}^{\infty} r^n < \infty.$$

Which thus concludes that the asymmetric random walk on $\mathbb{Z}$ is transient. We can use the similar notion to show the transience of 2-D random walks other than the symmetric case. [4]

A simple symmetric random walk on $\mathbb{Z}^3$: The transition probabilities of the simple symmetric random walk on $\mathbb{Z}^3$ are given by:

$$p_{ij} = \begin{cases} 1/6 & if \;\; |i-j| = 1 \\ 0 & otherwise \end{cases}$$

Thus the chain jumps to each of its nearest neighbours with equal probability. Suppose we start at 0. We can only return to 0 after an even number $2n$ of steps. Of these i up, i down, j north, j south and k east k west for some $i, j, k \geq 0$, with i+j+k=n. By seeing how this pans out in a combinatorial sense:

$$p_{00}^{(2n)} = \sum_{i,j,k \geq 0, i+j+k=n} \frac{(2n)!}{(i!\,j!\,k!\,)^2} \left(\frac{1}{6}\right)^{2n} = \binom{2n}{n} \left(\frac{1}{2}\right)^{2n} \sum_{i,j,k \geq 0, i+j+k=n} \binom{n}{i\;j\;k}^2 \left(\frac{1}{3}\right)^{2n}$$

Now,

$$\sum_{i,j,k \geq 0, i+j+k=n} \binom{n}{i\;j\;k}^2 \left(\frac{1}{3}\right)^{2n} = 1$$

the left-hand side being the total probability of all the ways of placing n balls randomly into three boxes. For the case where $n = 3m$, we have

$$\binom{n}{i\;j\;k} = \frac{n!}{i!\,j!\,k!} \leq \binom{n}{m\;m\;m}$$

$\forall\ i, j, k$, so, as $n \to \infty$,

$$p_{00}^{(2n)} \leq \binom{2n}{n}\left(\frac{1}{2}\right)^{2n}\binom{n}{m\ m\ m}\left(\frac{1}{3}\right)^{2n} \sim \frac{1}{2A^3}\left(\frac{6}{n}\right)^{(3/2)}$$

We can see on summation that it is a convergent series on comparison test with $\sum_{n=0}^{\infty} n^{-3/2}$. Thus we can say:

$$\sum_{n=0}^{\infty} p_{00}^{(n)} < \infty$$

Thus we have also shown that the simple symmetric random walk on $\mathbb{Z}^3$, is transient. [4]

Notion of Periodicity

Consider a discrete-time Markov chain $X = (X_0, X_1, X_2,)$ with (countable) state space S and transition probability matrix P. The period of a state $i \in S$ is

$$d(i) = \gcd\{n \in \mathbb{N} : P_{i,i}^n > 0\}$$

State i is aperiodic if $d(i) = 1$ and periodic if $d(i) > 1$. Thus, starting in i , the chain can return to i only at multiples of the period d , and d is the largest such integer. This will come in handy in our understanding and proofs relating to the ergodicity and limiting theorems of Markov Chains.

Proposition 4. *If $i \leftrightarrow j$ then, $d(i) = d(j)$. This implies that periodicity is a class property given by the communication class.*

Proof. Recall that since i, j communicate we can say $\exists\ n, m \in \mathbb{N}_0$ such that $p_{ij}^n > 0$ and $p_{ji}^m > 0$. From here we can say that:

$$p_{jj}^{n+m} \geq p_{ji}^m p_{ij}^n > 0$$

Hence, $d(j)|(m+n)$. Now suppose $\exists\ k > 0$ such that $p_{ii}^k > 0$. Here $d(i)|k$ Then we can say:

$$p_{jj}^{n+m+k} \geq p_{ji}^m p_{ii}^k p_{ij}^n > 0$$

So $d(j)|(n+m+k)$ and since it already divides $(m+n)$ we can see $d(j)|n$. So as period is gcd we can say, $d(i)|d(j)$. Now by reversing argument for i and j we can see $d(j) = d(i)$. □

Thus, the definitions of period, periodic, and aperiodic apply to equivalence classes as well as individual states. When the chain is irreducible, we can apply these terms to the entire chain.

1.4 Hitting Times and Probabilities

We shall write $P_i(A)$ for the conditional probability $P(A \in S|X_0 = i)$. Let $(X_n)_{n\in\mathbb{N}_0}$ be a Markov chain with transition matrix P. The hitting time of a subset A of state space S is the random variable H^A: $\Omega \to \mathbb{N}_0$ given by:-

$$H^A(w) = \inf\{n \geq 0 : X_n(\omega) \in A\}$$

where we agree that the infimum of the empty set $\emptyset$ is ∞. The probability starting from $i$that $(X_n)_{n\in\mathbb{N}_0}$ ever hits A is then called the hitting probability and is given by:-

$$h_i^A = P_i(H^A < \infty)$$

When A is a closed class, h_i^A is called the *absorption probability.* Remarkably, these quantities can be calculated explicitly by means of certain linear equations associated with the transition matrix P.

Theorem 1. *The vector of hitting probabilities $h^A = (h_i^A : i \in S)$is the minimal non-negative solution to the system of linear equations*

$$\begin{cases} h_i^A = 1, i \in A \\ h_i^A = \sum_{j\in S} p_{ij} h_j^A \end{cases} \tag{1.3}$$

(Minimality means that if $x = (x_i : i \in S)$ is another solution with $x_i \geq 0\ \forall\ i$, then $x_i \geq h_i\ \forall\ i$.)

Proof. If $X_0 = i \in A$, then $H^A = 0$, so $h_i^A = 1$. If $X_0 = i \notin A$, then $H^A \geq 1$, so by the Markov property:

$$P_i(H^A < \infty | X_1 = j) = P_j(H^A < \infty) = h_j^A$$

and thus,

$$h_i^A = P_i(H^A < \infty) = \sum_{j\in S} P_i(H^A < \infty, X_1 = j)$$

$$= \sum_{j\in S} P_i(H^A < \infty | X_1 = j) P_i(X_1 = j) = \sum_{j\in S} p_{ij} h_j^A$$

Suppose now that $x = (x_i : i \in S)$ is any solution to (1.3). Then $h_i^A = x_i = 1$ for $i \in A$. Suppose $i \notin A$, then

$$x_i = \sum_{j\in S} p_{ij} x_j = \sum_{j\in A} p_{ij} + \sum_{j\notin A} p_{ij} x_j$$

Now we can substitute for x_i to obtain:

$$x_i = \sum_{j\in A} p_{ij} + \sum_{j\notin A} p_{ij} \left(\sum_{k\in A} p_{jk} + \sum_{k\notin A} p_{jk} x_k \right)$$

$$= P_i(X_1 \in A) + P_i(X_1 \notin A, X_2 \in A) + \sum_{j\notin A} \sum_{k\notin A} p_{ij} p_{jk} x_k$$

By repeated substitution for X in the final term we obtain after n steps,

$$x_i = P_i(X_1 \in A) + ... + P_i(X_1 \notin A, ..., X_{n-1} \notin A, X_n \in A) + \sum_{j_1\notin A} ... \sum_{j_n\notin A} p_{ij_1} p_{j_1 j_2} ... p_{j_{n-1} j_n} x_{j_n}$$

Now if x is non-negative, so is the last term on the right, and the remaining terms sum up to $P_i(H^A \leq n)$. So $x_i \geq P_i((H^A \leq n)$ for all n and then, [5]

$$x_i \geq P_i(H^A \leq n) = P_i(H^A < \infty) = h_i$$

□

1.5 Long-run Proportions of time & Stationary Distributions

We will first state that any vector λ (we consider row vectors mainly) in which the entries are non-negative and add up-to 1, then λ is called a distribution. Now an idea which comes in handy for studying Markov Chains is what is called a Stationary Distribution. We say that λ is a stationary distribution if,

$$\lambda P = \lambda$$

That is it satisfies the equation:

$$\lambda_j = \sum_{i \in A} \lambda_i p_{ij} \tag{1.4}$$

The terms equilibrium and invariant are also used to mean the same. Let us now move on to the notion of long run proportions of time. Consider a random variable N_j which signifies number of transitions of Markov chains required to transition into state j. That is,

$$N_j = min\{n > 0 : X_n = j\}$$

Now we define the term m_j representing the expected number of transitions or average number of transitions that the Markov chain needs to return to state j while starting in state j. i.e:

$$m_j = E[N_j | X_0 = j]$$

Proposition 5. *If the Markov chain is irreducible and recurrent, then for any initial state*

$$\pi_j = 1/m_j \tag{1.5}$$

Where, π_j is the long run proportion of time spent by the chain in j. [1]

Proof. Let T_1 denote number of transitions until chain enter state j. Then let T_2 denote additional number after which it re-enters state j. We know T_1 is finite as recurrent. Also, for $T_2, T_3...$ they are i.i.d with mean m_j. Thus by definition of π_j we can state:

$$\begin{aligned}
\pi_j &= \lim_{n\to\infty} \frac{n}{\sum_{i=1}^{n} T_i} \\
&= \lim_{n\to\infty} \frac{1}{\frac{1}{n}\sum_{i=1}^{n} T_i} \\
&= \frac{1}{m_j}
\end{aligned}$$

Where the last equality follows as $\lim_{n\to\infty} T_1/n = 0$ and by the Strong Law of Large Numbers, $\lim_{n\to\infty} \frac{T_2+...+T_n}{n} = \lim_{n\to\infty} \frac{T_2+...+T_n}{n-1}\frac{n-1}{n} = m_j$. □

1.6 Convergence to Equilibrium, Limiting Theorems and Ergodicity

In this section we will state some very important theorems, firstly let us observe some facts about the stationary distribution:

Theorem 2. *Let $(X_n)_{n\geq 0}$ be $Markov(\lambda, P)$ (Markov chain with distribution λ and transition matrix P and suppose that λ is invariant for P. Then $(X_{m+n})_{n,m\geq 0}$ is also $Markov(\lambda, P)$.*

Proof. We can see if say that the initial distribution i.e. distribution of X_0 is stationary, say $\lambda_j = P(X_0 = j)$, then using eqn. (1.4),

$$P(X_1 = j) = \sum_{i\in A} P(X_1 = j|X_0 = i)P(X_0 = i)$$
$$P(X_1 = j) = \sum_{i\in A} \lambda_i p_{ij} = \lambda_j$$

By extending this using induction we see,

$$P(X_n = j) = \sum_{i\in A} P(X_n = j|X_{n-1} = i)P(X_{n-1} = i)$$
$$P(X_n = j) = \sum_{i\in A} \lambda_i p_{ij} = \lambda_j$$

Hence if initial stationary distribution then X_n also has a stationary distribution $\forall n \geq 0$. From this we easily see our proof follows. □

Now we will present the main theorem which is often called the Ergodic theorem for Markov Chains. [3]

Theorem 3. ***An irreducible, aperiodic Markov Chain belongs to the following two classes:***

i) Either the states are all transient or all null recurrent. In this case $P_{ij}^n \to 0$ as $n \to \infty$ and there is exists no stationary distribution.

ii) The states are all positive recurrent and thus

$$\lim_{n\to\infty} p_{ij}^n > 0$$

and in this case $\{\pi_j, j \in \mathbb{N}_0\}$ is the unique stationary distribution

I will not be including the proof due to the rigour it demands and this project is more focused on the applications of these ideas. However, main ideas used during the proof include the fact that m_j is finite only for (ii) while it is not so in the other case. The rest of the proof uses the idea of *Coupling* of two Markov chains which can be found in the first chapter of the book by J.R. Norris.[4]
This concludes our discussions on the convergence of Markov chains to the equilibrium or stationary distribution. I have also written a code which exhibits this convergence which can be found in *Appendix 2 Code 6.2* The case presented in part (ii) of theorem 3 is indeed what we call an Ergodic Markov Chain. Interestingly this result also satisfies if in (ii) the chain is periodic, [3] however then if the period is d, we arrive at the limiting result:

$$\lim_{n\to\infty} p_{ij}^{nd} = \frac{d}{m_j} = d\pi_j$$

1.7 Time Reversal

Suppose any distribution π satisfies the equations:

$$\pi_i p_{i,j} = \pi_j p_{ji} \tag{1.6}$$

$\forall\ i, j \in A$, these equations are called the detailed balance equation. [6]

Proposition 6. *Let P be a transition matrix of a Markov Chain with state space A. Any distribution π satisfying the detailed balance equation is a stationary distribution for P.*

Proof. Take sum over all j on both sides,

$$\begin{aligned} \sum_{j \in A} \pi_i p_{ij} &= \sum_{j \in A} \pi_j p_{ji} \\ \implies \pi_i &= \sum_{j \in A} \pi_j p_{ji} \end{aligned}$$

since, P is a stochastic matrix and thus it satisfies the conditions for π to be a stationary distribution. □

The *time reversal* of an irreducible Markov chain with transition matrix P and stationary distribution is the chain with matrix with elements:

$$p'_{ij} = \frac{\pi_j p_{ji}}{\pi_i} \tag{1.7}$$

By summing over all j we can easily see that this matrix is indeed stochastic. [1]

Proposition 7. *Let $(X_t)_{t \geq 0}$ be an irreducible Markov chain with transition matrix P and stationary distribution . Write $(X'_t)_{t \geq 0}$ for the time-reversed chain with transition matrix P'. Then π is stationary for P', and for any $x_0, ..., x_t \in A$ we have,*

$$P_\pi\{X_0 = x_0, ..., X_t = x_t\} = P'_\pi\{X'_0 = x_t, ..., X'_t = x_0\}$$

Proof. To show, both of these trajectories are equal we will use equation (1.7).

$$\begin{aligned} P_\pi\{X_0 = x_0, ..., X_t = x_t\} &= \pi_{x_0} p_{x_0 x_1} ... p_{x_{t-1} x_t} \\ &= \pi_{x_t} p'_{x_t x_{t-1}} ... p'_{x_1 x_0} \\ &= P'_\pi\{X'_0 = x_t, ..., X'_t = x_0\} \end{aligned}$$

Where we use the fact that, $p_{x_{i-1} x_i} = \frac{\pi_{x_i} p_{x_i x_{i-1}}}{\pi_{x_{i-1}}}$ from equation (1.7). Hence it is proved. □

Now we observe the fact that if a chain with transition matrix P is reversible, then $P' = P$. It is worth noting that this is an obvious necessary condition for time reversibility since a transition from i to j going backward in time is equivalent to a transition from j to i going forward in time.

1.8 The Poisson Process

Definition 1. *A counting process is a stochastic process defined as $\{N(t), t \geq 0\}$, where the random variable N denotes the number of "events" that occur in time t.*

From the definition [1] we see that a counting process has the following properties;

- $N(t) \geq 0$
- $N(t) \in \mathbb{Z}^+$
- If $s < t$, then $N(s) < N(t)$.
- Now if we have $s < t$ then, $N(t) - N(s)$ denotes the number of events taking place in the time between t and s

Now we consider two very important "behaviours" of these types of processes.
Firstly we state that a process is said to have *"independent increments"* if the numbers of events that occur in disjoint time intervals are independent. That is, the number of events occurring by s say $N(s)$ is independent of the number of events occurring between s and $s + t$ i.e $N(t) - N(s)$.
Next, we say a process has *stationary increments* if the distribution of number of events occurring in any time interval depends only on the time interval. In other words, the process has stationary increments if the number of events in the interval $(s, s+t)$ has the same distribution for all $s > 0$. [2] Now before we go further we define an useful type of function.

Definition 2. *A function $f(.) : \mathbb{R} \to \mathbb{R}$ is said to be $o(h)$ if,*

$$\lim_{h \to 0} \frac{f(h)}{h} = 0$$

Few properties of these are stated without proofs (which are trivial)

- If $f(.)$, $g(.)$ are $o(h)$, then so is $f(.) + o(.)$
- If $f(.)$ is $o(h)$, then so is $g(h) = cf(h)$ where $c \in \mathbb{R}$

Thus from this we can infer that any finite linear combination of $o(h)$ functions are in fact also $o(h)$.
After having defined these, we put forward 2 equivalent definitions of the Poisson Process:

Definition 3. *A counting process $\{N(t), t \geq 0\}$ is called a Poisson process with rate λ, $\lambda > 0$ if:*

- $N(0) = 0$.
- *It has independent increments.*
- *The number of events in any interval of length t is Poisson distributed with mean λt. That is, for all $s, t \geq 0$:*

$$P\{N(t+s) - N(s) = n\} = e^{-\lambda t} \frac{(\lambda t)^n}{n!} \tag{1.8}$$

We can simply see that the RHS of (1.8) is independent of s implying also the fact that it has stationary increments.

Now for an alternate definition:

Definition 4. *A counting process* $\{N(t), t \geq 0\}$ *is called a Poisson process with rate* λ, $\lambda > 0$ *if:*

- $N(0) = 0$.
- *It has independent increments and stationary increments*
- $P\{N(h) = 1\} = \lambda h + o(h)$
- $P\{N(h) \geq 2\} = o(h)$

To Prove: Definition 3 and Definition 4 are equivalent

Proof. Let us start by showing how definition 3 $\implies$ definition 4. Firstly we notice that points 1 and 2 are same for both the definitions, so what we have to show is that the last two are equivalent to the third point. Let us take a small time h, and try to find the probability from definition 3 that the only 1 event occurs in that interval using eqn (1.8)

$$P\{N(h) = 1\} = \lambda h e^{-\lambda h}$$
$$P\{N(h) = 1\} = \lambda h \left(1 - \lambda h + \frac{(\lambda h)^2}{2!}\right)$$
$$P\{N(h) = 1\} = \lambda h - (\lambda h)^2 + \frac{(\lambda h)^2}{2!}$$

By considering the series of the second powers and above terms we see that these are in fact all $o(h)$ as $h \to 0$. So hence we can see we have proved,

$$P\{N(h) = 1\} = \lambda h + o(h)$$

Now by considering this and definition 1 we get,

$$P\{N(h) = 0\} = 1 - \lambda h + o(h)$$

So now using the notion of sum of all probability as 1 we indeed get:

$$P\{N(h) \geq 2\} = o(h)$$

Hence, we have proved that Definition 3 $\implies$ 4. Now to prove the opposite fix a $u \geq 0$ and see the Moment generating function or the Laplace Transform of the random variable $N(t)$.

$$g(t) = E[e^{-uN(t)}]$$

Now we need to find what is $g(t + h)$ where $h \to 0$ to find the derivative.

$$g(t + h) = E[e^{-uN(t)}]E[e^{-u\{N(t+h)-N(t)\}}]$$
$$g(t + h) = g(t)E[e^{-u\{N(t+h)-N(t)\}}]$$
$$g(t + h) = g(t)E[e^{-uN(h)}]$$

Where we have used properties of independent and stationary increments. Now we have to find out the expectation value which turns out to be from points 3 and 4,

$$E[e^{-uN(h)}] = 1 - \lambda h + o(h) + e^{-u}(\lambda h + o(h)) + o(h)$$
$$E[e^{-uN(h)}] = 1 - \lambda h + o(h) + e^{-u}\lambda h$$

Now using this we can use first principle to find out the derivative thus,

$$\begin{aligned}
\frac{g(t+h) - g(t)}{h} &= \frac{g(t)\{1 - \lambda h + o(h) + e^{-u}\lambda h - 1\}}{h} \\
\lim_{h \to 0} \frac{g(t+h) - g(t)}{h} &= g(t)\lambda\{e^{-u} - 1\} \\
\frac{g'(t)}{g(t)} &= \lambda\{e^{-u} - 1\} \\
\ln g(t) &= \lambda t\{e^{-u} - 1\} + c
\end{aligned}$$

Now, the integration constant c becomes 0 when we put the boundary condition that $g(0) = 1$. Thus we get, [1]

$$g(t) = e^{\lambda t\{e^{-u} - 1\}}$$

Which is in fact the Moment generating function for a Poisson random variable and since choice of t was arbitrary we can see how this implies that it has a Poisson distribution. Thus 4 $\implies$ 3 and we are done. □

Inter-arrival and Waiting Times

Definition 5. *Consider a Poisson process, and let us denote the time of the first event by T_1. Further, for $n > 1$, let T_n denote the elapsed time between the $(n-1)st$ and the nth event. The sequence $\{T_n, n = 1, 2, ...\}$ is called the sequence of inter-arrival times.*

We shall now determine the distribution of the T_n. [4] To do so, we first note that the event $(T_1 > t)$ takes place if and only if no events of the Poisson process occur in the interval $[0, t]$ and thus,

$$P(T_1 > t) = P(N(t) = 0) = e^{-\lambda t}$$

Hence, T_1 has an exponential distribution with mean $1/\lambda$. Now,

$$P(T_2 > t) = E[P(T_2 > t|T_1)]$$

However we see,

$$\begin{aligned}
P(T_2 > t|T_1 = s) &= P(0 \; events \; in(s, s+t]|T_1 = s) \\
&= P(0 \; events \; in(s, s+t]) \\
&= e^{-\lambda t}
\end{aligned}$$

Where we have used arguments of independent and stationary increments and this we can see a very important fact if we repeat this argument further.

Proposition 8. *$T_n, n = 1, 2, ...,$ are independent identically distributed exponential random variables having mean $1/\lambda$.*

Consider a Poisson process $\{N(t), t \geq 0\}$ having rate λ, and suppose that each time an event occurs it is classified as either a type I or a type II event. Suppose further that each event is classified as a type I event with probability p or a type II event with probability $1-p$, independently of all other events.
Let $N_1(t)$ and $N_2(t)$ denote respectively the number of type I and type II events occurring in $[0, t]$. Note that $N(t) = N_1(t) + N_2(t)$.

Proposition 9. *$\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ are both Poisson processes having respective rates λp and $\lambda(1-p)$. Furthermore, the two processes are independent.*

Proof. It is very easy to see that these are in-fact Poisson processes just by conditioning on 1 event occurring and using Definition 4.
Now, because the probability of a type I event in the interval from t to $t+h$ is independent of all that occurs in intervals that do not overlap $(t, t+h)$, it is independent of knowledge of when type II events occur, showing that the two Poisson processes are independent [1].

We can present a more mathematical proof for the independence as follows: Let $N(t)t \geq 0$ be a Poisson Process with parameter λ . Suppose that each event is type 1 with probability p and type 2 with probability $1-p$. Let $N_1(t)t \geq 0$ the number of type 1 events up until time t and $N_2(t)t \geq 0$ the number of type 2 events up until time t. We have the Poisson process:

$$N(t) = N_1(t) + N_2(t)$$

Now, the joint probability is given by:

$$P[N_1(t) = k, N_2(t) = m] = \sum_{n=0}^{\infty} P[N_1(t) = k, N_2(t) = m | N(t) = n] P[N(t) = n]$$

Since we know that when $n \neq m + k$,

$$P[N_1(t) = k, N_2(t) = m | N(t) = n] = 0$$

Thus, we can write that:

$$P[N_1(t) = k, N_2(t) = m] = P[N_1(t) = k, N_2(t) = m | N(t) = m + k] P[N(t) = m + k]$$

But using definitions of Poisson Processes,

$$P[N_1(t) = k, N_2(t) = m] = P[N_1(t) = k, N_2(t) = m | N(t) = m + k] e^{-\lambda t} \frac{(\lambda t)^{m+k}}{(m+k)!}$$

Now, given that $k+m$ events occurred, since each event has probability p of being a type 1 event and probability $1-p$ of being a type 2 event, it follows that:

$$P[N_1(t) = k, N_2(t) = m | N(t) = m + k] = \binom{m+k}{k} p^k (! - p)^m$$

Thus,

$$\begin{aligned} P[N_1(t) = k, N_2(t) = m] &= \frac{(k+m)!}{m!\, k!} p^k (! - p)^m e^{-\lambda t} \frac{(\lambda t)^{m+k}}{(m+k)!} \\ &= e^{-\lambda p t} \frac{(\lambda p t)^k}{k!} e^{-\lambda(1-p)t} \frac{(\lambda p t)^m}{m!} \end{aligned}$$

Thus we see that $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ are both **independent Poisson processes** having respective rates λp and $\lambda(1-p)$. □

1.9 A Brief Introduction to Continuous Time Markov Chains

Let X_t be a family of random variables, parameterized by $t \in \mathbb{R}^+$, with values in a discrete set $A(e.g., \mathbb{Z})$. To extend the notion of Markov chain to that of a continuous time Markov chain one naturally requires:

$$P[X_{s+t} = j | X_s = i, X_{s_n} = i_n, , X_{s_1} = i_1] = P[X_{s+t} = j | X_s = i] \tag{1.9}$$

In other words, a continuous-time Markov chain is a stochastic process having the Markov property that the conditional distribution of the future $X_(t+s)$ given the present $X_(s)$ and the past, depends only on the present and is independent of the past. If, in addition,

$$P\{X(t+s) = j | X(s) = i\}$$

is independent of s, then the continuous-time Markov chain is said to have stationary or homogeneous transition probabilities. [5]

By virtue of the Markov Property, if we let T_i denote the amount of time that the process stays in state i before making a transition into a different state then,

$$P\{T_i > s + t | T_i > s\} = PT_i > t$$

Which is essentially the memory-less property and since among continuous random variables, only the exponential distribution has this property we can conclude that: T_i is **memory-less** and must thus be **exponentially distributed**.

Now we can say that any continuous time Markov chain is a stochastic process having the properties that each time it enters state i,

- the amount of time it spends in that state before making a transition into a different state is exponentially distributed with mean, say, $1/\lambda_i$, and
- when the process leaves state i, it next enters state j with some probability, say, p_{ij}. Of course, the p_{ij} must satisfy:

$$p_{ii} = 0 \quad \forall \ i$$
$$\sum_{j \in A} p_{ij} = 1 \quad \forall \ i$$

Where it follows from the fact that if it re-enters i, it implies the state of *"not leaving"* and thus it will not follow an exponential distribution as postulated. [1]

Birth and Death Processes

A birth and death process is a type of Markov process that models various real-world phenomena where entities (such as individuals in a population) are subject to births and deaths over time. The key features of a birth and death process include:

- **States:** The process moves between non-negative integer states, each representing the number of entities.
- **Births:** Transitions from state n to state $n+1$ occur at a rate λ_n.
- **Deaths:** Transitions from state n to state $n-1$ occur at a rate μ_n.

That is, whenever there are n persons in the system, then the time until the next arrival is exponentially distributed with mean $1/\lambda_n$ and is independent of the time until the next departure, which is itself exponentially distributed with mean $1/\mu_n$. [1]

Chapter 2

Applications of Markov Chain Modelling

In this chapter we will look at how we can use Markov Chain Modelling to work out some problems which may not at first seem to be so intuitive.

2.1 Ultimate Instability of Aloha Protocol

Consider a communications facility in which the numbers of messages arriving during each of the time periods $n = 1, 2, ...$ are independent and identically distributed random variables. Let $a_i = P\{i \text{ arrivals}\}$, and suppose that $a_0 + a_1 < 1$. Each arriving message will transmit at the end of the period in which it arrives. If exactly one message is transmitted, then the transmission is successful and the message leaves the system. However, if at any time two or more messages simultaneously transmit, then a collision is deemed to occur and these messages remain in the system. Once a message is involved in a collision it will, independently of all else, transmit at the end of each additional period with probability p - the so called Aloha protocol (because it was first instituted at the University of Hawaii). Now to show that such a system is asymptotically unstable in the sense that the number of successful transmissions will, with probability 1, be finite.[1]

Let X_n denote the number of messages in the facility at the beginning of the nth period, and note that $X_n, n \geq 0$ is a Markov chain. Now for $k \geq 0$ define the indicator variables I_k by:

$$I_k = \begin{cases} 1, \text{if the first time that the chain departs state } k \text{ it directly goes to state } k-1 \\ 0, \text{ otherwise} \end{cases}$$

and let it be 0 if the system is never in state $k, k \geq 0$.(For instance,if the successive states are $0, 1, 3, 4, ...$, then $I_3 = 0$ since when the chain first departs state 3 it goes to state 4;

whereas, if they are $0, 3, 3, 2, ...$, then $I_3 = 1$ since this time it goes to state 2.) Now,

$$\begin{aligned} E\left[\sum_{k=0}^{\infty} I_k\right] &= \sum_{k=0}^{\infty} E\left[I_k\right] \\ &= \sum_{k=0}^{\infty} P\left[I_k = 1\right] \\ &\leq \sum_{k=0}^{\infty} P\left[I_k | k \textit{ is ever visited}\right] \end{aligned}$$

Now, $P\left[I_k | k \text{ is ever visited}\right]$ is the probability that when state k is departed the next state is $k-1$. That is, it is the conditional probability that a transition from k is to $k-1$ given that it is not back into k, and so,

$$P\left[I_k | k \textit{ is ever visited}\right] = \frac{P_{k,k-1}}{1 - P_{k,k}}$$

Also we can see,

$$\begin{aligned} P_{k,k-1} &= a_0 kp(1-p)^{k-1}, \\ P_{k,k} &= a_0[1 - kp(1-p)^{k-1}] + a_1(1-p)^k \end{aligned}$$

which is seen by noting that if there are k messages present on the beginning of a day, then (a) there will be $k-1$ at the beginning of the next day if there are no new messages that day and exactly one of the k messages transmits; and (b) there will be k at the beginning of the next day if either:

- there are no new messages and it is not the case that exactly one of the existing k messages transmits, or
- there is exactly one new message (which automatically transmits) and none of the other k messages transmits.

Substitution of the preceding leads us to the expression:

$$E\left[\sum_{k=0}^{\infty} I_k\right] \leq \sum_{k=0}^{\infty} \frac{a_0 kp(1-p)^{k-1}}{1 - a_0[1 - kp(1-p)^{k-1}] + a_1(1-p)^k}$$

where the convergence follows by noting that when k is large the denominator of the expression in the preceding sum converges to $1 - a_0$ and so the convergence or divergence of the sum is determined by whether or not the sum of the terms in the numerator converge and $\sum_{k=0}^{\infty} kp(1-p)^{k-1} < \infty$
$E\left[\sum_{k=0}^{\infty} I_k\right] < \infty$, which implies that $\sum_{k=0}^{\infty} I_k < \infty$ with probability 1 (for if there was a positive probability that $\sum_{k=0}^{\infty} I_k$ could be ∞, then its mean would be ∞). Hence, with probability 1, there will be only a finite number of states that are initially departed via a successful transmission; or equivalently, there will be some finite integer N such that whenever there are N or more messages in the system, there will never again be a successful transmission.
From this it follows that, with probability 1, there will only be a finite number of successful transmissions, thus exhibiting that in the asymptotic case it is unstable.

2.2 The Gambler's Ruin

A unique formulation of the one-dimensional Random Walk problem is the "Gambler's Ruin" Problem. The problem of Gambler's Ruin can be traced back to the $17th$ century, a period when the foundations of probability theory were being laid by mathematicians like Blaise Pascal and Pierre de Fermat. Their correspondence and work on problems posed by gamblers, particularly the Chevalier de Méré, led to the birth of formal probability theory. Later in 1657, Huygens wrote *"De ratiociniis in ludo aleae"* (On Reasoning in Games of Chance), which is considered one of the first books on probability theory. While it did not explicitly solve the Gambler's Ruin problem, it laid important groundwork for understanding random processes and gambling problems.

Formulation of the Problem:

The Gambler's ruin problem can be formulated as follows: [1] Consider a gambler who at each play of the game has probability p of winning one unit and probability $q = 1 - p$ of losing one unit. Assuming that successive plays of the game are independent, starting with i units, we can frame two questions of interest:

- What is the probability that the gambler's fortune will reach N before reaching 0?

- The expected number of bets until ruin i.e. 0 wealth or reaching the target wealth N.

If we let X_n denote the player's fortune at time n, then the process $X_n, n = 0, 1, 2, ...$ is a Markov chain with transition probabilities:

$$p_{00} = p_{NN} = 1,$$
$$p_{i,i+1} = p = 1 - p_{i,i-1},\ i = 1, 2, ..., N - 1$$

These states 0 and N are called absorbing states. This Markov chain has three classes, namely, $\{0\}$, $\{1, 2, ..., N - 1\}$, and $\{N\}$; the first and third class being recurrent and the second transient. Since each transient state is visited only finitely often, it follows that, after some finite amount of time, the gambler will either attain his goal of N or go broke. Let $P_i, i = 0, 1, ...N$ denote the probability that starting in i, the gambler's fortune will eventually reach N. We can easily note that $P_0 = 0$ and $P_N = 1$. Now by conditioning on the first transition out of i we can obtain,

$$P_i = pP_{i+1} + qP_{i-1}$$

and since, $p + q = 1$,

$$pP_i + qP_i = pP_{i+1} + qP_{i-1}$$

So now we can rearrange and write that,

$$P_{i+1} - P_i = \frac{q}{p}(P_{i-1} - P_i)$$

for $i = 1, ..., N - 1$. Hence, since $P_0 = 0$, we obtain from the preceding line that by repetative argument,

$$P_N - P_{N-1} = \left(\frac{q}{p}\right)^{N-1} P_1$$

Adding the first $i-1$ of these equations yields:

$$P_i - P_1 = P_1 \left[\left(\frac{q}{p}\right) + \left(\frac{q}{p}\right)^2 + ... + \left(\frac{q}{p}\right)^{i-1} \right]$$

Now using the fact that $P_N = 1$ and by using formula for sum of finite terms of a geometric progression we get,

$$P_i = \begin{cases} \frac{1-(q/p)^i}{1-(q/p)^N}, \ if \ p \neq \frac{1}{2} \\ \frac{i}{N}, \ if \ p = \frac{1}{2} \end{cases}$$

Now as $N \to \infty$ for some i which is finite we can say that:

$$P_i = \begin{cases} 1 - (q/p)^i, \ if \ p > \frac{1}{2} \\ 0, \ if \ p \leq \frac{1}{2} \end{cases}$$

Thus, if $p > 1/2$, there is a positive probability that the gambler's fortune will increase indefinitely, while if $p \leq 1/2$, the gambler will,with probability 1, go broke against an infinitely rich adversary. This has been shown in Figure 2.1 which was generated using the *code 6.3* in Appendix 2.

As for the case of Expected duration, we can look up the theory for solving linear homogeneous and non-homogeneous difference equations which leads us to the solution. This can be found in detail in notes by Matthew Aldrige [7].

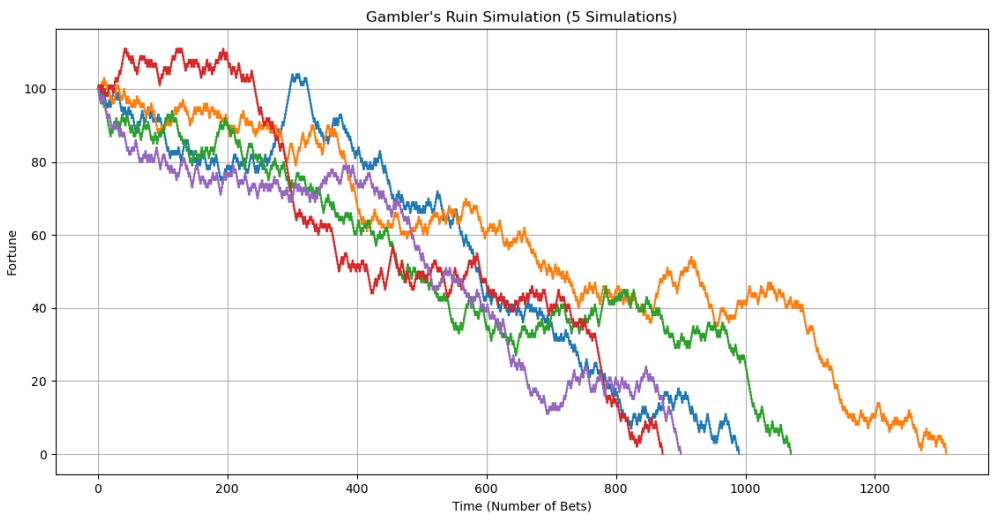


Figure 2.1: Gambler's Ruin starting with 100 units with $N = 1000$ for 2000 bets

Chapter 3

Markov Chain Monte Carlo Methods

3.1 Introduction

The Monte Carlo Method was primarily developed Developed during the 1940s by Stanislaw Ulam, John von Neumann, and others for solving complex physical problems, particularly in nuclear physics. It is a class of computer algorithms which use repeated random sampling to approach simulations. By the 1950s and 1960s, researchers realized that combining Markov chains with Monte Carlo methods could improve the efficiency and applicability of Monte Carlo simulations, particularly for problems involving high-dimensional spaces. The Metropolis algorithm, developed by Nicholas Metropolis [10] and colleagues, was one of the first MCMC methods. It uses a Markov chain to sample from a probability distribution by accepting or rejecting proposed moves based on a specific criterion (the Metropolis criterion). W.K. Hastings [12] generalized the Metropolis algorithm, resulting in the Metropolis-Hastings algorithm, which allows for more flexible proposal distributions and broader applicability.

3.2 Metropolis-Hastings Algorithm & Gibbs Sampler

The Metropolis-Hastings algorithm is a popular MCMC method. It constructs a Markov chain using a proposal distribution $q(x'|x)$ to generate candidate states. It was devised by Metropolis et. al. [10] and Hastings [12]. Further work was presented on sampling using MCMC and calculation of marginal densities by Gelfand and others [11].

Algorithm

Given a current state x_t, the next state x_{t+1} is generated as follows:

1. Propose a new state x' from the proposal distribution $q(x'|x_t)$.
2. Compute the acceptance ratio:
$$\alpha = \min\left(1, \frac{\pi(x')q(x_t|x')}{\pi(x_t)q(x'|x_t)}\right).$$
3. Accept the new state with probability α:
   - If $u \leq \alpha$, where $u \sim \text{Uniform}(0,1)$, set $x_{t+1} = x'$.
   - Otherwise, set $x_{t+1} = x_t$.

Proof of Detailed Balance

The Metropolis-Hastings algorithm satisfies the detailed balance condition, which ensures that the Markov chain has the desired stationary distribution $\pi(x)$.

Detailed Balance Condition A Markov chain with transition probability matrix P and stationary distribution π satisfies detailed balance if:

$$\pi(x)P(x \to x') = \pi(x')P(x' \to x) \quad \forall x, x' \in \mathcal{S}.$$

Proof Let $P(x \to x')$ be the transition probability from state x to state x'. In the Metropolis-Hastings algorithm:

$$P(x \to x') = q(x'|x)\alpha,$$

where $\alpha = \min\left(1, \frac{\pi(x')q(x|x')}{\pi(x)q(x'|x)}\right)$.

Consider the product $\pi(x)P(x \to x')$:

$$\pi(x)P(x \to x') = \pi(x)q(x'|x)\min\left(1, \frac{\pi(x')q(x|x')}{\pi(x)q(x'|x)}\right).$$

Similarly,

$$\pi(x')P(x' \to x) = \pi(x')q(x|x')\min\left(1, \frac{\pi(x)q(x'|x)}{\pi(x')q(x|x')}\right).$$

By examining the two cases within the minimum function, it can be shown that:

$$\pi(x)P(x \to x') = \pi(x')P(x' \to x),$$

which proves detailed balance.

Gibbs Sampling

Gibbs sampling is a special case of the Metropolis-Hastings algorithm where the proposal distribution is chosen such that the candidate state is always accepted.

Algorithm

Given a joint distribution $\pi(x_1, x_2, \ldots, x_d)$, Gibbs sampling updates one variable at a time while keeping the others fixed. For each variable x_i, sample from its conditional distribution:

$$x_i^{(t+1)} \sim \pi(x_i|x_1^{(t+1)}, x_2^{(t+1)}, \ldots, x_{i-1}^{(t+1)}, x_{i+1}^{(t)}, \ldots, x_d^{(t)}).$$

Proof of Convergence

Gibbs sampling converges to the target distribution $\pi(x)$ because it satisfies the detailed balance condition. Since each variable is sampled from its exact conditional distribution, the resulting Markov chain has $\pi(x)$ as its stationary distribution.

Applications of MCMC

MCMC methods are widely used in various fields such as:

- **Bayesian Inference**: MCMC is used to sample from posterior distributions in Bayesian statistics.
- **Statistical Physics**: Simulation of physical systems at equilibrium like the Ising model which will be exhibited in the next section.
- **Machine Learning**: Parameter estimation and model selection in complex models.

3.3 Applications in the Ising Model

Development of Theory

The Ising model is concerned with the physics of phase transitions, which occur when a small change in a parameter such as temperature or pressure causes a large-scale, qualitative change in the state of a system. Initially proposed by Wilhelm Lenz in 1920 and solved in one dimension by his student Ernst Ising in 1925, serves as a cornerstone in statistical mechanics and condensed matter physics. Ising's one-dimensional solution showed no phase transition at finite temperature, but Lars Onsager's exact solution of the two-dimensional Ising model in 1944 [15] demonstrated a phase transition, revolutionizing the understanding of critical phenomena. Subsequent work by *Rudolf Peierls (1936)* [19] and later rigorous results by *C. N. Yang (1952)* [16] and *T. D. Lee and C. N. Yang (1952)* [16] further solidified its importance. The Ising model has since been extended to various lattices, dimensions, and even quantum versions, proving indispensable in studying magnetism, lattice gas models, and even sociophysics.

MCMC Methods to Simulate Ising Model

Implementing the Metropolis-Hastings Algorithm is actually very simple with regards to the Ising Model. Here we choose the targeted distribution to be the Boltzmann Distribution i.e. the probability being in a certain spin configuration say x is given by:

$$P(x) = e^{\frac{H(x)}{k_B T}}$$

where $H(x)$ is basically the Hamiltonian which gives the energy of the system.
The Ising model describes a system of spins $\sigma_i = \pm 1$ arranged on a lattice, interacting with nearest neighbors via an energy function given by:

$$H = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j - h \sum_i \sigma_i$$

where J denotes the interaction strength between spins, $\langle i, j \rangle$ denotes nearest neighbors, h is an external magnetic field, and σ_i are the spin variables. This model serves as a fundamental tool in statistical mechanics and has been extensively studied for its phase transition properties, critical phenomena, and applications in diverse fields of physics and beyond. A great introductory but technical paper for Ising Model is the paper by *Barry A. Cipra* (1987) [20].

Ising Model Simulation and Animation Algorithm

Here we have given the pseudo-code for simulating the Ising Model using the Metropolis-Hastings Algorithm. Codes are included in *Appendix 2 code 6.1.*

Initialization and Simulation

1. **Initialize Parameters:**
   - N: Lattice size (N x N)
   - J: Interaction strength
   - h: External magnetic field
   - T: Temperature
   - $\beta = \frac{1}{T}$: Inverse temperature
   - `steps`: Number of simulation steps
2. **Initialize Functions:**
   - `initialize_lattice(N)`: Create an $N \times N$ lattice with random spins (+1 or -1).
   - `calculate_energy(lattice, J, h)`: Calculate the energy of the lattice configuration using the Ising model Hamiltonian.
   - `metropolis_step(lattice, J, h,`β: Perform one Metropolis-Hastings update step to evolve the lattice configuration.
   - `simulate_ising(N, J, h,` β`, steps)`: Simulate the Ising model over `steps` time steps, tracking energy evolution and lattice configurations.
3. **Simulation Execution:**
   - Call `simulate_ising(N, J, h,` β`, steps)` to obtain `energy_list` and `lattice_list`.
   - `energy_list`: List containing the energy values at each simulation step.
   - `lattice_list`: List of lattice configurations at each simulation step.

Animation Setup

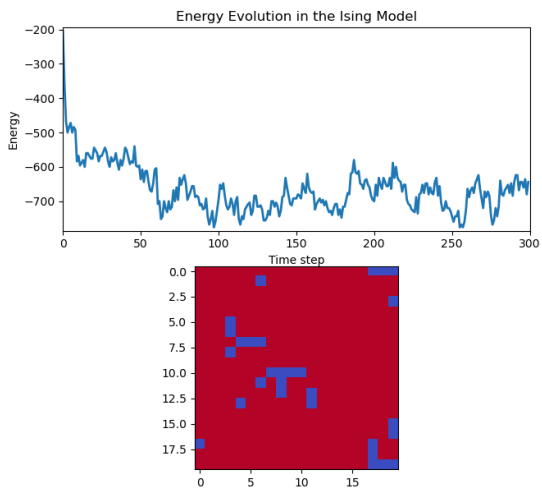
1. **Initialize Plotting:**
   - Create a figure with two subplots:
     - `ax_energy`: Plot for energy evolution over time steps.
     - `ax_lattice`: Plot for visualizing lattice configurations.
   - Set initial plot parameters for `ax_energy`.
2. **Initial Plotting of Energy Evolution:**
   - Plot the initial energy evolution using `line_energy`.
   - Set labels and title for the energy plot.
3. **Initialize Lattice Visualization:**

- Create an initial image plot (`im`) of the lattice configuration using `lattice_list[0]`.
- Use a colormap (e.g., 'coolwarm') to visualize spin orientations.

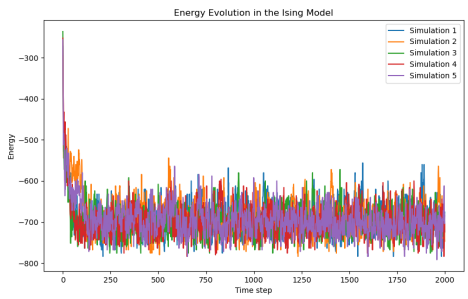
4. **Define Update Function for Animation (`update(frame)`):**
   - Update the image plot (`im`) with the lattice configuration from `lattice_list[frame]`.
   - Update the energy plot (`line_energy`) with energy values up to the current time step (`frame`).
   - Return the updated objects (`im`, `line_energy`) for animation.
5. **Animate the Simulation (`animation.FuncAnimation`):**
   - Use `animation.FuncAnimation` to animate the lattice evolution and energy plot updates.
   - Set the interval between frames (`interval`) to control animation speed.
   - Specify `blit=True` for efficient updating of the plot elements.
6. **Display the Animation:**
   - Show the animated plot using `plt.show()`.

Results and Figures

I have got outputs regarding how the system behaves over a period of time steps t=300 here and plots showing the process of energy minimization and lattice visualisation as given in Figure 3.1a. If you want to see the running animation please **click here**.



(a) Energy Evolution and Visualisation for a Ising Model Lattice with h=0, 20x20 2-D Lattice, J=1, T=2.0



(b) Energy Minimization Curves for $n = 5$ Simulations at $h = 0$, $T = 2.0$

Figure 3.1: Results from Simulations

The other code I have written shows how we can see the patterns for energy simulations minimizing for 5 different iterations of the algorithm which can be seen in Figure 3.1b.

Chapter 4

Brownian Motion & Wiener Processes

Definition 6. *A stochastic process $X(t), t \geq 0$ is said to be a Brownian motion process if:*

- $X(0) = 0$.
- $X(t), t \geq 0$ *has stationary and independent increments.*
- *for every $t > 0$, $X(t)$ is normally distributed with mean 0 and variance $\sigma^2 t$.*

Brownian motion is often modeled as having increments that are independent and identically distributed. This means that the changes in position over small time intervals are independent of each other and have the same statistical properties.This leads to the Wiener process as it is called also, by essence of Central Limit Theorem to have Gaussian Increments. We have utilized this and used the `numpy_random.randn` function to code a Brownian Simulation for a system of particles whose program is available in *Appendix 2 code 6.4* along with a video of the simulation for 50 particles which can be found **here**. This function is used as it helps us generate a random sample (or samples) from the "standard normal" distribution and the "2" in the code signifies the dimension of each particle. [5]

Brownian Motion as the Limit of a Symmetric Random Walk:

Let us start by considering the symmetric random walk, which in each time unit is equally likely to take a unit step either to the left or to the right. That is, it is a Markov chain with $P_{i,i+1} = \frac{1}{2} = P_{i,i-1}$, $i \in \mathbb{Z}$. Now suppose that we speed up this process by taking smaller and smaller steps in smaller and smaller time intervals. If we now go to the limit in the right manner what we obtain is Brownian motion. [6]
Suppose that each Δt time unit we take a step of size Δx either to the left or the right with equal probabilities. If we let $X(t)$ denote the position at time t then,

$$X(t) = \Delta x(X_1 + \ + X_{[t/\Delta t]})$$

where, $[t/\Delta t]$ denotes largest integer less than or equal to $t/\Delta t$. Here,

$$X_i = \begin{cases} +1 \ if \ ith \ step \ of \ length \ \Delta x \ is \ to \ the \ right. \\ -1, \ if \ it \ is \ to \ the \ left. \end{cases}$$

and the X_i are assumed independent with:

$$P\{X_i = 1\} = P\{X_i = -1\} = \frac{1}{2}$$

As $E[X_i] = 0$, $Var(X_i) = E[X_i^2] = 1$, we see from the previous equations that:

$$E[X(t)] = 0$$
$$Var(X(t)) = (\Delta x)^2 \left[\frac{t}{\Delta t}\right]$$

We shall now let Δx and Δt go to 0. However, we must do it in a way such that the resulting limiting process is nontrivial (for instance, if we let $\Delta x = \Delta t$ and let $\Delta t \to 0$, then from the preceding we see that E[X(t)] and Var(X(t)) would both converge to 0 and thus X(t) would equal 0 with probability 1). If we let $\Delta x = \sigma\sqrt{\Delta t}$ for some positive constant σ then from previous equations we see that as $\Delta t \to 0$,

$$E[X(t)] = 0$$
$$Var(X(t)) \to \sigma^2 t$$

We now list some intuitive properties of this limiting process obtained:

- $X(t)$ is normal with mean 0 and variance $\sigma^2 t$. In addition, because the changes of value of the random walk in non-overlapping time intervals are independent.
- $X(t), t \geq 0$ has independent increments, in that $\forall\ t1 < t2 < ... < tn$

  $$X(t_n) - X(t_{n-1}), X(t_{n-1}) - X(t_{n-2}), ..., X(t_2) - X(t_1), X(t_1)$$

  are independent. Finally, because the distribution of the change in position of the random walk over any time interval depends only on the length of that interval, it would appear that:
- $X(t), t \geq 0$ has stationary increments, in that the distribution of $X(t+s) - X(t)$ does not depend on t. Thus we have arrived at the formal definition. [1]

Brownian Motion also has an interesting Physical Theory which was given by Einstein in his seminal work on the topic. A good paper *"On the Theory of the Brownian Motion"* by *G. E. Uhlenbeck and L. S. Ornstein (1930)* [17] gives the subject a thorough review.

Chapter 5

Prospective Future Research & Conclusion

Markov Processes lead to a variety of interdisciplinary topics of research in fields as varied as Finance, Quantum systems, Protein folding, Genetics and so on. Modelling of Transport Phenomena using Exclusion Processes which have been used since solved by *Derrida et. al. (1993)* [8] [9] to analyse and study not only processes in Statistical Mechanics but also in biological systems and traffic flows [18].

Due to my personal interest in Physics I would like to explore how Markov Processes can be utilized to learn dynamics of Complex Open Quantum Systems. Some open questions in this field include:

- How do Markovian and non-Markovian processes affect the dynamics of entanglement and other quantum correlations in open quantum systems?
- How can different stochastic unravelings of the master equation be used to gain insights into the behavior of open quantum systems?
- How do quantum jump processes and quantum trajectories, which are inherently Markovian, contribute to our understanding of measurement and dynamics in open quantum systems?

And, of course many more. Another particularly interesting question which I will look to work upon is the process of developing algorithms for efficient generation and also identification of periodic Markov transition matrices. This may/may not be original work but will be interesting to look at.

I will conclude the discussions by emphasising how invaluable a resource Markov Processes and Stochastic Processes in general, are for our understanding of the Physical World and how we should strive to integrate these models to gather information on Nature.

Chapter 6

Appendix

6.1 Appendix 1: Historical Aspect

Early Developments

Andrey Markov (1906):
The concept of Markov chains was introduced by Russian mathematician Andrey Markov in 1906. He studied sequences of random variables where the future state depends only on the present state and not on the past states. This property is known as the Markov property.

First Applications

Markov initially applied his theory to study the distribution of vowels and consonants in the works of the Russian poet Alexander Pushkin, demonstrating the utility of Markov chains in linguistic analysis.

Formalization and Expansion

Early 20th Century: The mathematical theory of Markov chains was further developed and formalized by various mathematicians. Kolmogorov's work in the 1930s on the foundations of probability theory provided a rigorous mathematical framework for studying stochastic processes, including Markov chains.

Eugen Slutsky (1925): Slutsky used Markov chains in his work on stochastic processes and time series analysis, contributing to the understanding of their applications in economics and other fields.

Claude Shannon (1948): Shannon applied Markov chains in his groundbreaking work on information theory. He used them to model the statistical properties of English text, which played a crucial role in the development of communication theory and cryptography.

Erdős and Rényi (1960): Mathematicians Paul Erdős and Alfréd Rényi studied the connectivity properties of random graphs using Markov chains, which laid the groundwork for the field of random graph theory.

Modern Developments

Renewed Interest in 1970s and 1980s: The advent of computers and advances in computational methods led to a resurgence in the study and application of Markov chains. Applications expanded to fields like queuing theory, statistical mechanics, and genetics.

Markov Chain Monte Carlo (MCMC): The development of MCMC methods in the 1950s and their popularization in the 1990s revolutionized statistical inference. These methods use Markov chains to sample from complex probability distributions, enabling applications in Bayesian statistics, machine learning, and computational biology.

6.2 Appendix 2: Codes

For detailed output videos and to use these codes, one can access it from this GitHub Repository.

1. Ising Model using MCMC Method

```python

import numpy as np
import matplotlib.pyplot as plt
import matplotlib.animation as animation

def initialize_lattice(N):
    return np.random.choice([-1, 1], size=(N, N))

def calculate_energy(lattice, J, h):
    N = lattice.shape[0]
    energy = 0
    for i in range(N):
        for j in range(N):
            S = lattice[i, j]
            neighbors = lattice[(i+1)%N, j] + lattice[i, (j+1)%N] + lattice[(i-1)%N, j] + lattice[i, (j-1)%N]
            energy += -J * S * neighbors - h * S
    return energy / 2  # Each pair counted twice

def metropolis_step(lattice, J, h, beta):
    N = lattice.shape[0]
    for _ in range(N**2):
        i = np.random.randint(0, N)
        j = np.random.randint(0, N)
        S = lattice[i, j]
        dE = 2 * S * (J * (lattice[(i+1)%N, j] + lattice[i, (j+1)%N] + lattice[(i-1)%N, j] + lattice[i, (j-1)%N]) + h)
        if dE < 0 or np.random.rand() < np.exp(-beta * dE):
            lattice[i, j] = -S
    return lattice

def simulate_ising(N, J, h, beta, steps):
    lattice = initialize_lattice(N)
    energy_list = []
    lattice_list = []
```

```
    for step in range(steps):
        lattice = metropolis_step(lattice, J, h, beta)
        energy = calculate_energy(lattice, J, h)
        energy_list.append(energy)
        lattice_list.append(lattice.copy())
    return energy_list, lattice_list

# Parameters
N = 20              # Lattice size (N x N)
J = 1.0             # Interaction strength
h = 0      # External magnetic field
T = 2         # Temperature
beta = 1.0 / T      # Inverse temperature
steps = 300       # Number of simulation steps
interval = 50       # Interval between frames in milliseconds


energy_list, lattice_list = simulate_ising(N, J, h, beta, steps)


fig, axs = plt.subplots(2, 1, figsize=(8, 8))
ax_energy = axs[0]
ax_lattice = axs[1]

line_energy, = ax_energy.plot([], [], lw=2)
ax_energy.set_xlim(0, steps)
ax_energy.set_ylim(min(energy_list) - 10, max(energy_list) + 10)
ax_energy.set_xlabel('Time step')
ax_energy.set_ylabel('Energy')
ax_energy.set_title('Energy Evolution in the Ising Model')


im = ax_lattice.imshow(lattice_list[0], cmap='coolwarm', animated=True)

def update(frame):
    im.set_array(lattice_list[frame])
    line_energy.set_data(range(frame + 1), energy_list[:frame + 1])
    return [im, line_energy]


ani = animation.FuncAnimation(fig, update, frames=range(steps), interval=
    interval, blit=True)

plt.tight_layout()
plt.show()
```

Listing 6.1: Ising Model Energy Evolution & 2-D Lattice Evolution

2. Markov Transition Matrix Multiplication & exhibition of Stationary Distribution

```
def get_matrix_input(n):
    matrix = []
    print(f"Enter the {n}x{n} matrix row by row (space-separated values):")
    for i in range(n):
        row = list(map(float, input(f"Row {i + 1}: ").split()))
        if len(row) != n:
```

```
            raise ValueError("Invalid row length")
        matrix.append(row)
    return matrix

def check_row_sums(matrix):
    for row in matrix:
        if not abs(sum(row) - 1.0) < 1e-6:
            return False
    return True

def matrix_multiply(A, B):
    n = len(A)
    result = [[0] * n for _ in range(n)]
    for i in range(n):
        for j in range(n):
            for k in range(n):
                result[i][j] += A[i][k] * B[k][j]
    return result

def matrix_power(matrix, power):
    result = [[1 if i == j else 0 for j in range(len(matrix))] for i in
        range(len(matrix))]
    base = matrix
    while power:
        if power % 2 == 1:
            result = matrix_multiply(result, base)
        base = matrix_multiply(base, base)
        power //= 2
    return result

def find_stationary_vectors(matrix):
    n = len(matrix)
    s = [1/n] * n
    for _ in range(100):
        new_s = [0] * n
        for j in range(n):
            for i in range(n):
                new_s[j] += s[i] * matrix[i][j]
        s = new_s
    return s

def main():
    n = int(input("Enter the size of the square matrix: "))
    matrix = get_matrix_input(n)

    if not check_row_sums(matrix):
        print("Each row of the matrix does not sum to 1. Exiting.")
        return

    P10 = matrix_power(matrix, 10)
    print("Matrix multiplied by itself 10 times:")
    for row in P10:
        print(" ".join(map(str, row)))

    P20 = matrix_power(matrix, 20)
    print("Matrix multiplied by itself 20 times:")
    for row in P20:
        print(" ".join(map(str, row)))
```

```python

    stationary_vector = find_stationary_vectors(matrix)
    print("Row vector s such that sP = s:")
    print(" ".join(map(str, stationary_vector)))

if __name__ == "__main__":
    main()
```

Listing 6.2: Exhibition of Stationary Distribution

3. Gambler's Ruin Simulation

```python
import numpy as np
import matplotlib.pyplot as plt

def gamblers_ruin_simulation(initial_fortune, target_fortune, p, max_steps
    =1000):
    fortune = initial_fortune
    fortunes = [fortune]

    for _ in range(max_steps):
        if fortune == 0 or fortune == target_fortune:
            break
        bet_result = np.random.choice([-1, 1], p=[1-p, p])
        fortune += bet_result
        fortunes.append(fortune)

    return fortunes


n = int(input("Enter the number of simulations: "))
initial_fortune = int(input("Enter the initial fortune: "))
target_fortune = int(input("Enter the target fortune: "))
max_steps = int(input("Enter the maximum number of steps: "))
p = float(input("Enter the probability of winning each bet (less than 0.5
    to be against the gambler): "))


all_fortunes = []
for _ in range(n):
    fortunes = gamblers_ruin_simulation(initial_fortune, target_fortune, p,
        max_steps)
    all_fortunes.append(fortunes)


plt.figure(figsize=(12, 8))
for fortunes in all_fortunes:
    plt.plot(fortunes, drawstyle='steps-post')

plt.xlabel('Time (Number of Bets)')
plt.ylabel('Fortune')
plt.title(f'Gambler\'s Ruin Simulation ({n} Simulations)')
plt.grid(True)
plt.show()
```

Listing 6.3: Exhibition of Stationary Distribution

4. Brownian Motion Simulation of a Particle System

```python
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.animation import FuncAnimation

num_particles = 50
num_steps = 1000
step_size = 0.1

positions = np.zeros((num_particles, 2))

fig, ax = plt.subplots()
ax.set_xlim(-10, 10)
ax.set_ylim(-10, 10)
particles, = ax.plot([], [], 'bo', ms=6)

def update(frame):
    global positions
    positions += np.random.randn(num_particles, 2) * step_size
    particles.set_data(positions[:, 0], positions[:, 1])
    return particles,

ani = FuncAnimation(fig, update, frames=num_steps, interval=50, blit=True)

plt.title('Brownian Motion Simulation')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.show()
```

Listing 6.4: Brownian Simulation

References

[1] **Sheldon Ross (2014)** *Introduction to Probability Models (Eleventh Edition*, Elsevier Inc.

[2] **Sheldon Ross (2010)** *A First Course in Probability (Ninth Edition)*, Pearson.

[3] **Sheldon Ross (1995)** *Stochastic Processes (Second Edition)*, John Wiley & Sons, Inc.

[4] **J.R. Norris (1997)** *Markov Chains (First Edition*, Cambridge University Press

[5] **Prof. Hao Wu (2015)** *18.445 Introduction to Stochastic Processes*, MIT OpenCourseWare.

[6] **Prof. Gordan Zitkovic (2019)** *M362M Introduction to Stochastic Processes*, University of Texas at Austin.

[7] **Prof. Matthew Aldrige (2021)** *MATH2750 Introduction to Markov Processes*, University of Leeds.

[8] **B. Derrida et. al. (1993)** *Exact solution of a one dimensional asymmetric exclusion model using a matrix formulation*, J. Phys. A, 26.

[9] **B. Derrida et. al. (1993)** *Exact diffusion constant for one dimensional asymmetric exclusion models*, J. Phys. A, 26.

[10] **Metropolis et. al. (1953)** *Equation of State Calculations by Fast Computing Machines*, J. Chem. Phys. 21

[11] **Gelfland et. al. (1990)** *Sampling-Based Approaches to Calculating Marginal Densities*, J. Amer. Statist. Assoc., 85

[12] **W.K. Hastings (1970)** *Monte Carlo Sampling Methods Using Markov Chains and Their Applications*, Biometrika 57. Oxford University Press

[13] **Hossein Pishro-Nik (2014)** *Introduction to probability, statistics, and random processes*, University of Massachusetts.

[14] **Brush, S. G. (1967)** *History of the Lenz-Ising Model*, Rev. Mod. Phys. 39

[15] **Lars Onsager (1944)** *Crystal Statistics. I. A Two-Dimensional Model with an Order-Disorder Transition*, Phys. Rev. 65

[16] **C.N. Yang (1952)** *The Spontaneous Magnetization of a Two-Dimensional Ising Model*, Phys. Rev. 85

[17] **G. E. Uhlenbeck & L. S. Ornstein (1930)** *On the Theory of the Brownian Motion*, Phys. Rev. 36

[18] **D. Chowdhury et. al. (2005)** *Physics of transport and traffic phenomena in biology: from molecular motors and cells to organisms*, Phys. of Life Rev. 2

[19] **Rudolf Peierls (1936)** *On Ising's model of ferromagnetism*, Math. Proceed. of the Cambridge Phil. Soc. , 32

[20] **Barry A. Cipra (1987)** *An Introduction to the Ising Model*, American Math. Monthly 94